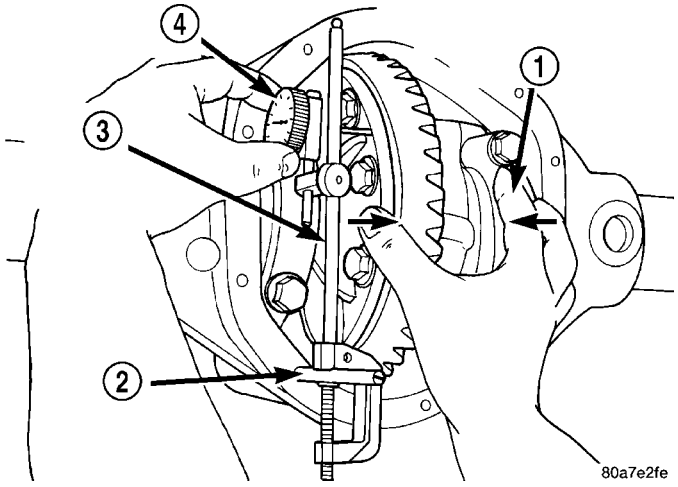

Fig. 19 DIAL INDICATOR LOCATION

1 - FORCE DIFFERENTIAL CASE TO PINION GEAR SIDE
2 - PILOT STUD
3 - DIAL INDICATOR
4 - ZERO DIAL INDICATOR FACE

(11) Push and hold differential case to ring gear side of the housing and record the dial indicator reading (Fig. 20).

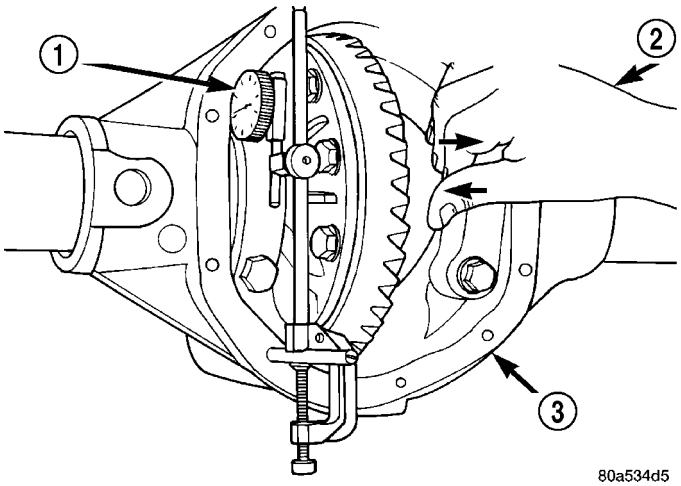

Fig. 20 READ DIAL INDICATOR

1 - READ DIAL INDICATOR
2 - DIFFERENTIAL CASE TO RING GEAR SIDE
3 - DIFFERENTIAL HOUSING

(12) Add 0.008 in. (0.2 mm) to the zero end play total. This new total represents the thickness of shims to compress or preload the new bearings when the differential is installed.

(13) Rotate dial indicator out of the way on the pilot stud.

(14) Remove differential case and dummy bearings from the housing.

(15) Install the pinion gear in the housing. Install the pinion yoke and establish the correct pinion rotating torque.

(16) Install differential case and dummy bearings D-348 in the housing (without shims), install bearing caps and tighten bolts snug.

(17) Seat ring gear side dummy bearing (Fig. 17).

(18) Position the dial indicator plunger on a flat surface between the ring gear bolt heads (Fig. 18).

(19) Push and hold differential case toward pinion gear and zero the dial indicator (Fig. 21).

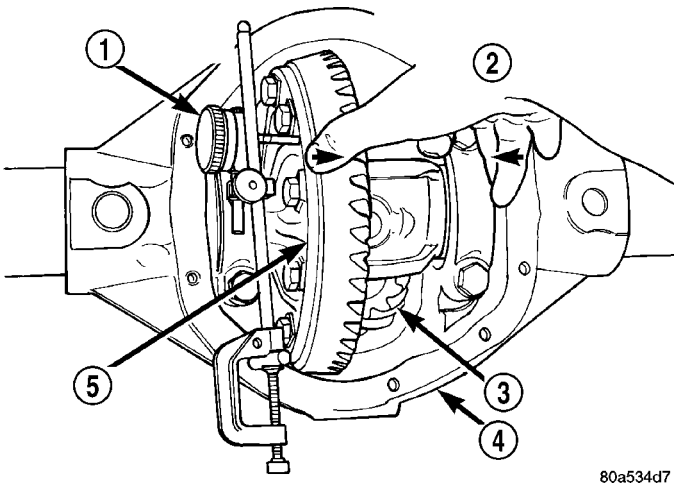

Fig. 21 ZERO DIAL INDICATOR

1 - ZERO DIAL INDICATOR FACE
2 - FORCE DIFFERENTIAL CASE TO PINION GEAR SIDE
3 - PINION GEAR
4 - DIFFERENTIAL HOUSING
5 - DIFFERENTIAL CASE

(20) Push and hold differential case to ring gear side of the housing and record dial indicator reading (Fig. 22).

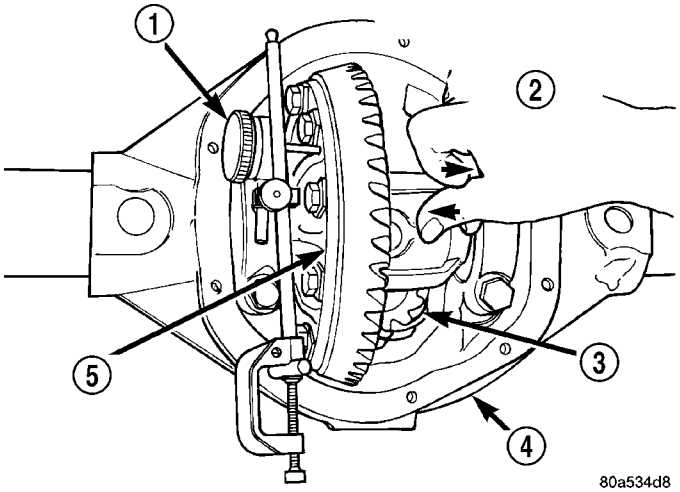

Fig. 22 DIFFERENTIAL CASE RING GEAR SIDE

1 - READ DIAL INDICATOR
2 - FORCE DIFFERENTIAL CASE TO RING GEAR SIDE
3 - PINION GEAR
4 - DIFFERENTIAL HOUSING
5 - DIFFERENTIAL CASE

(21) Subtract 0.05 mm (0.002 in.) from the dial indicator reading to compensate for backlash between ring and pinion gears. This total is the thickness shim required to achieve proper backlash.

(22) Subtract the backlash shim thickness from the total preload shim thickness. The remainder is